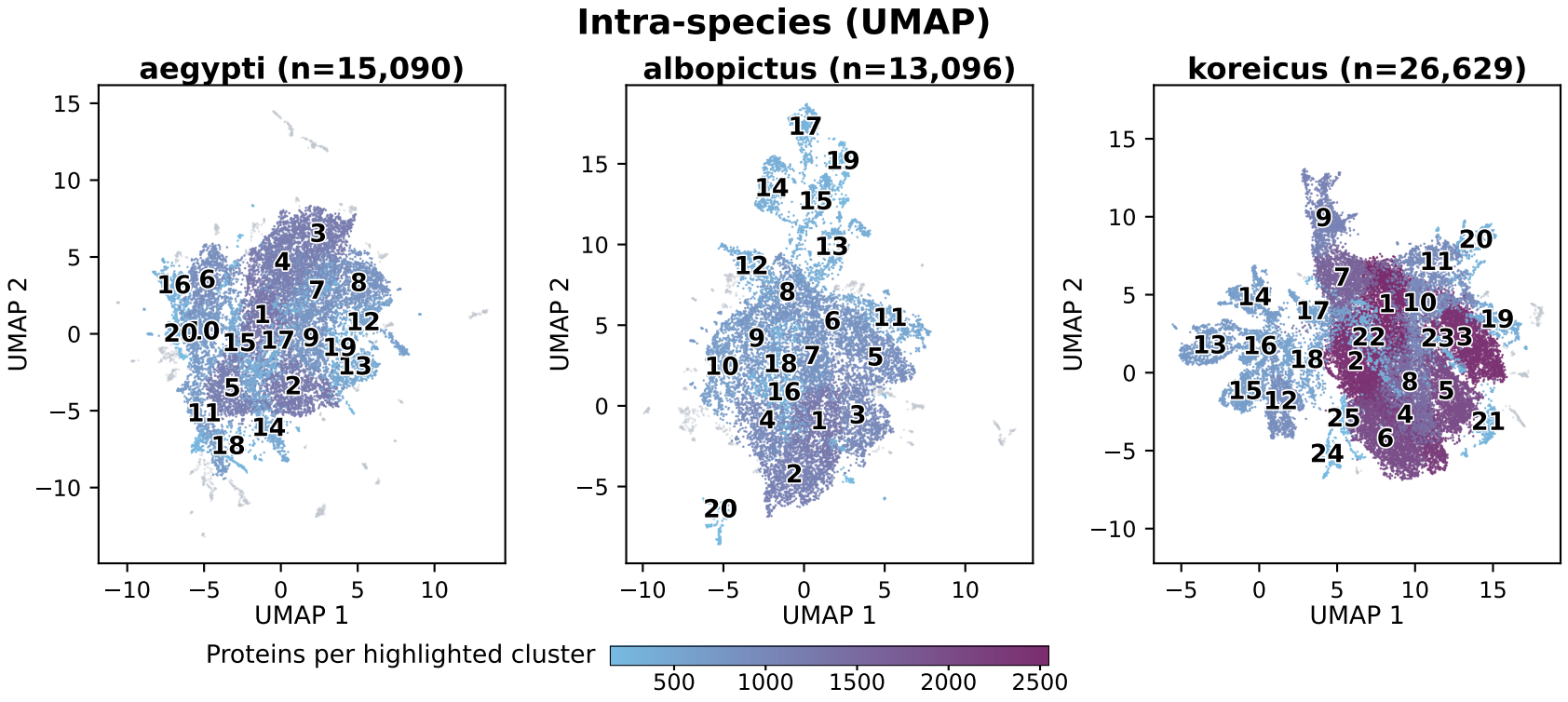
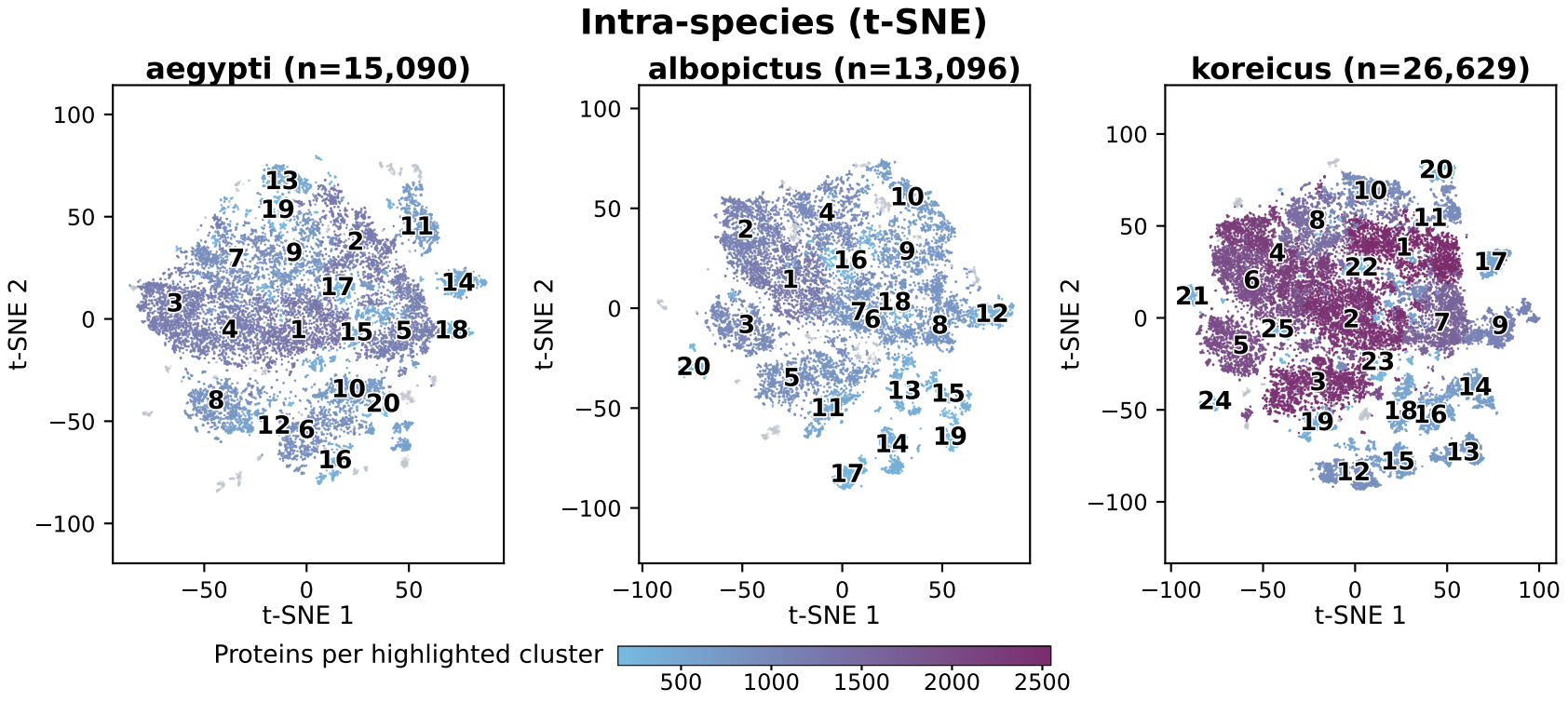
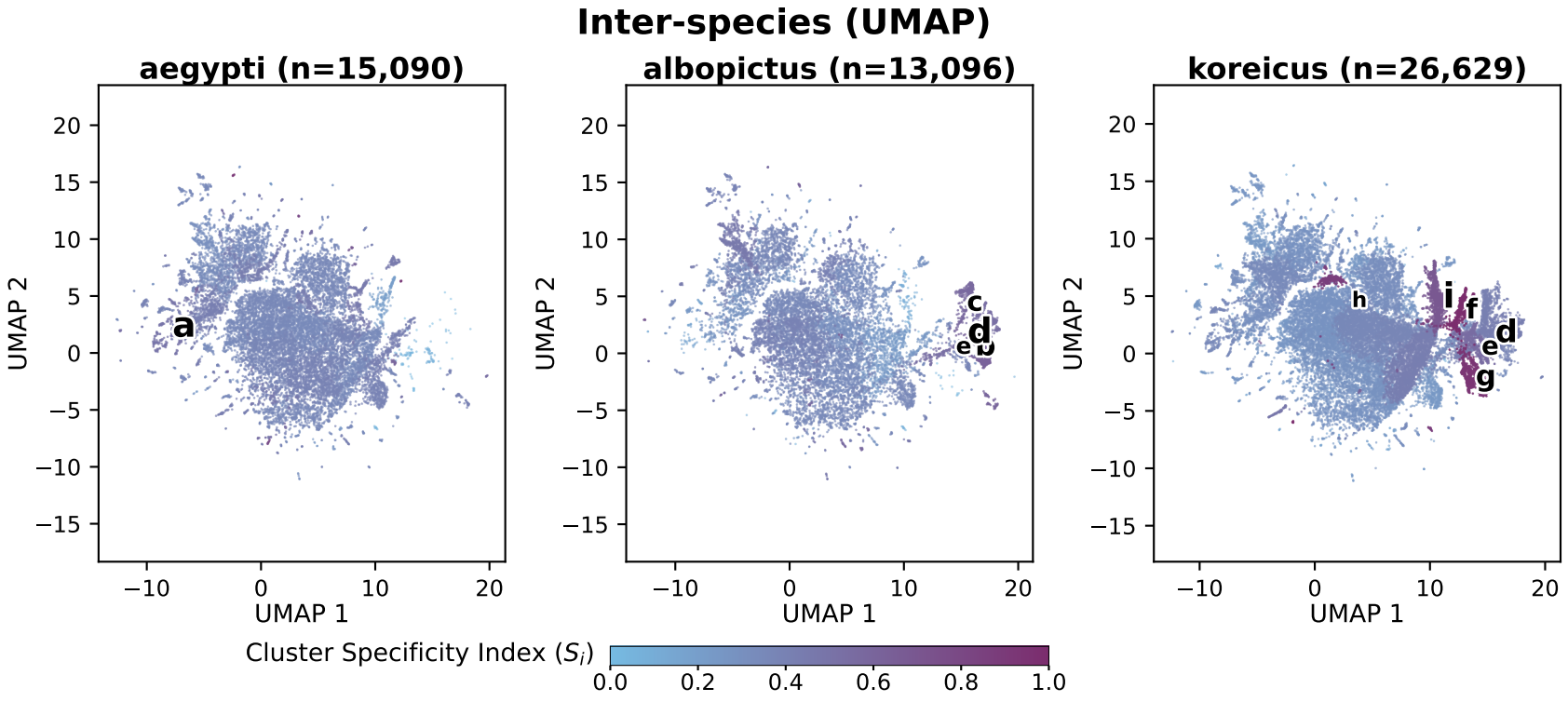
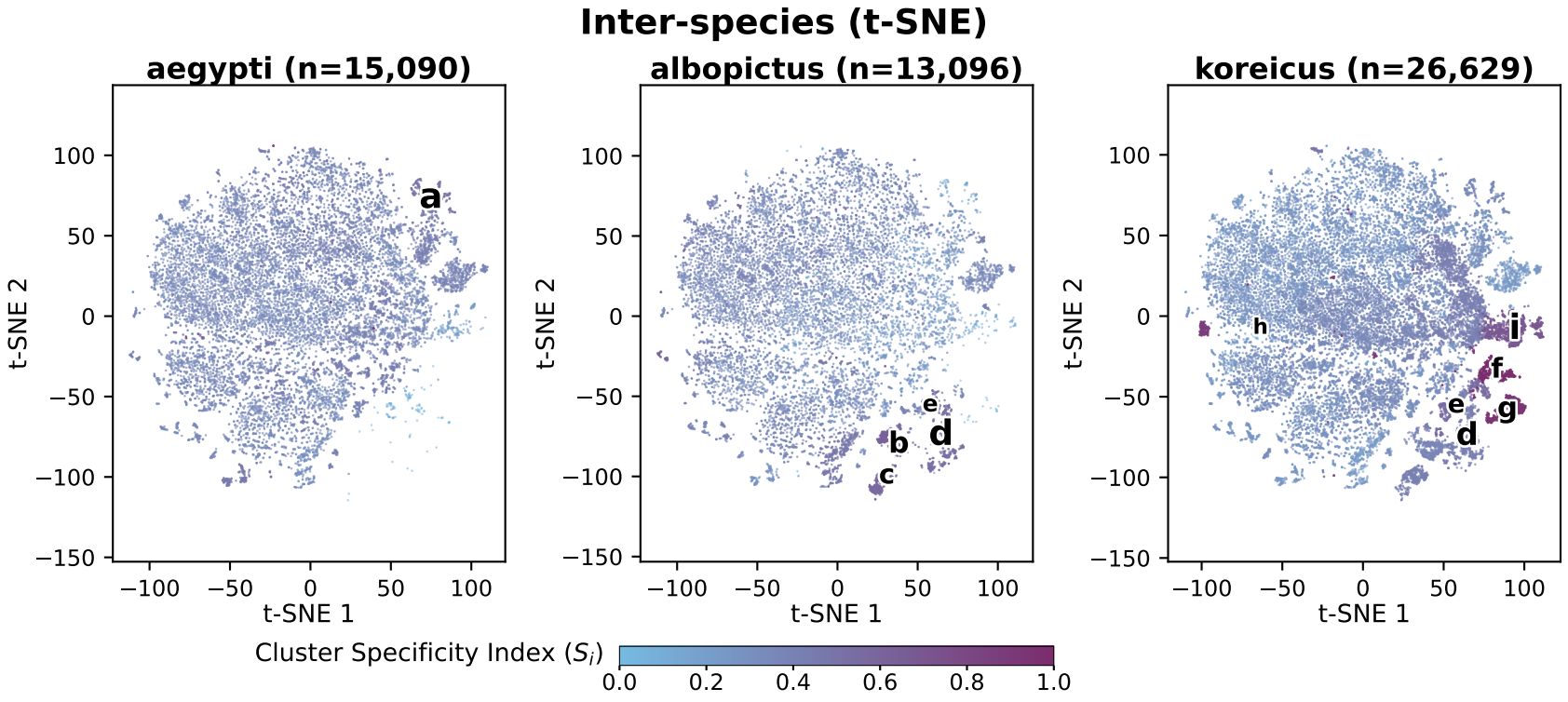
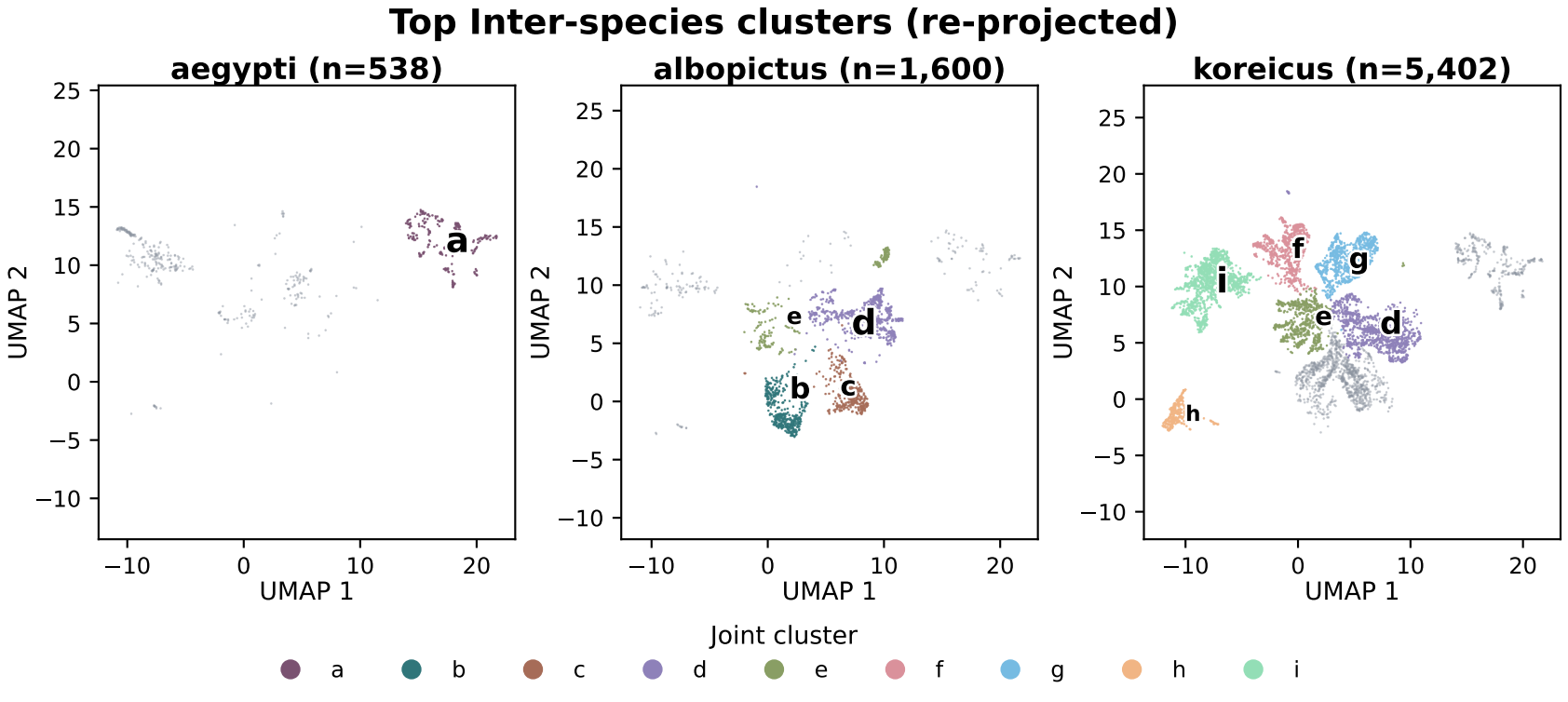
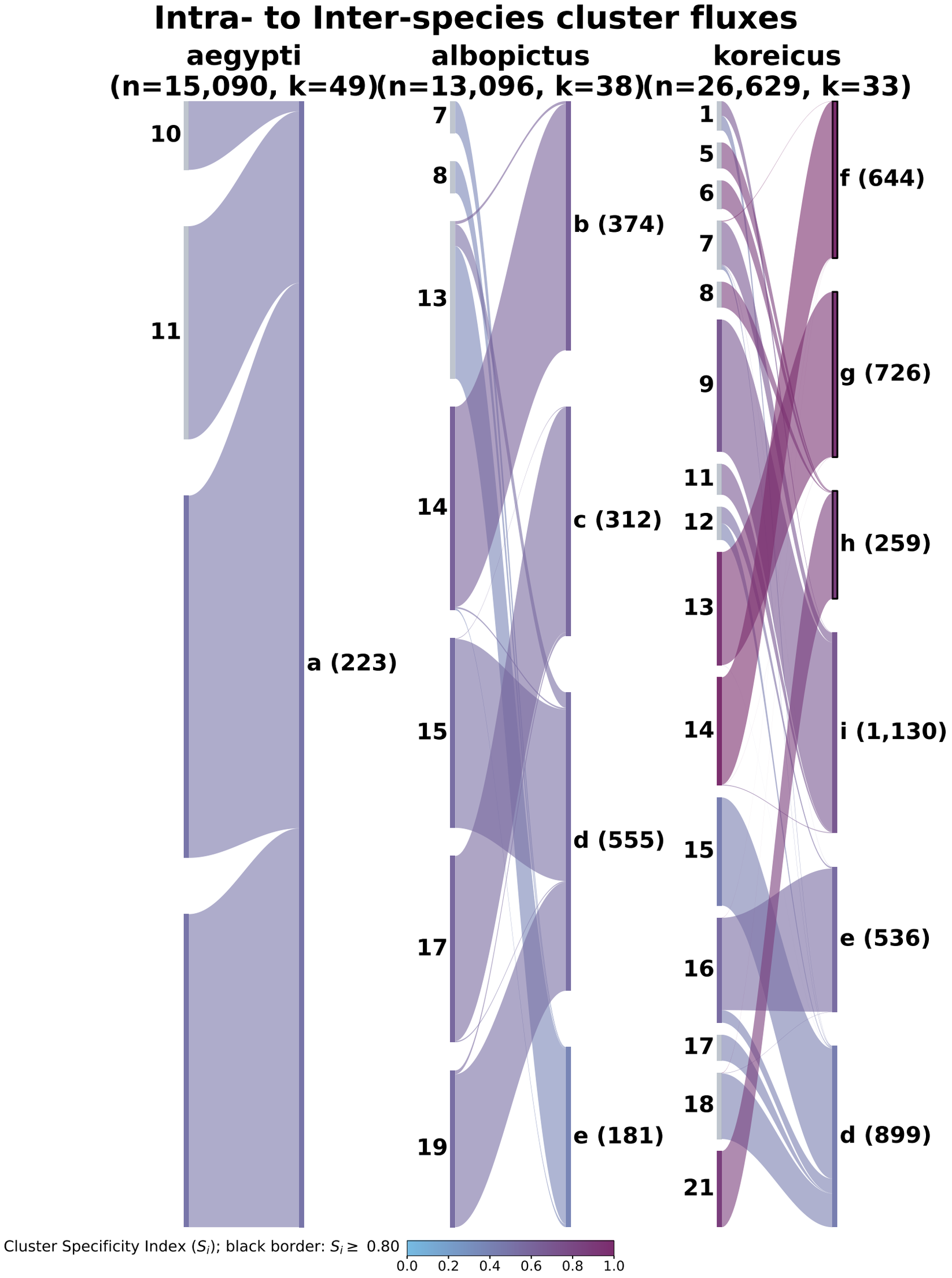

Intra-species (UMAP)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
UMAP 1
UMAP 2
Proteins per highlighted cluster
500
1000
1500
2000
2500

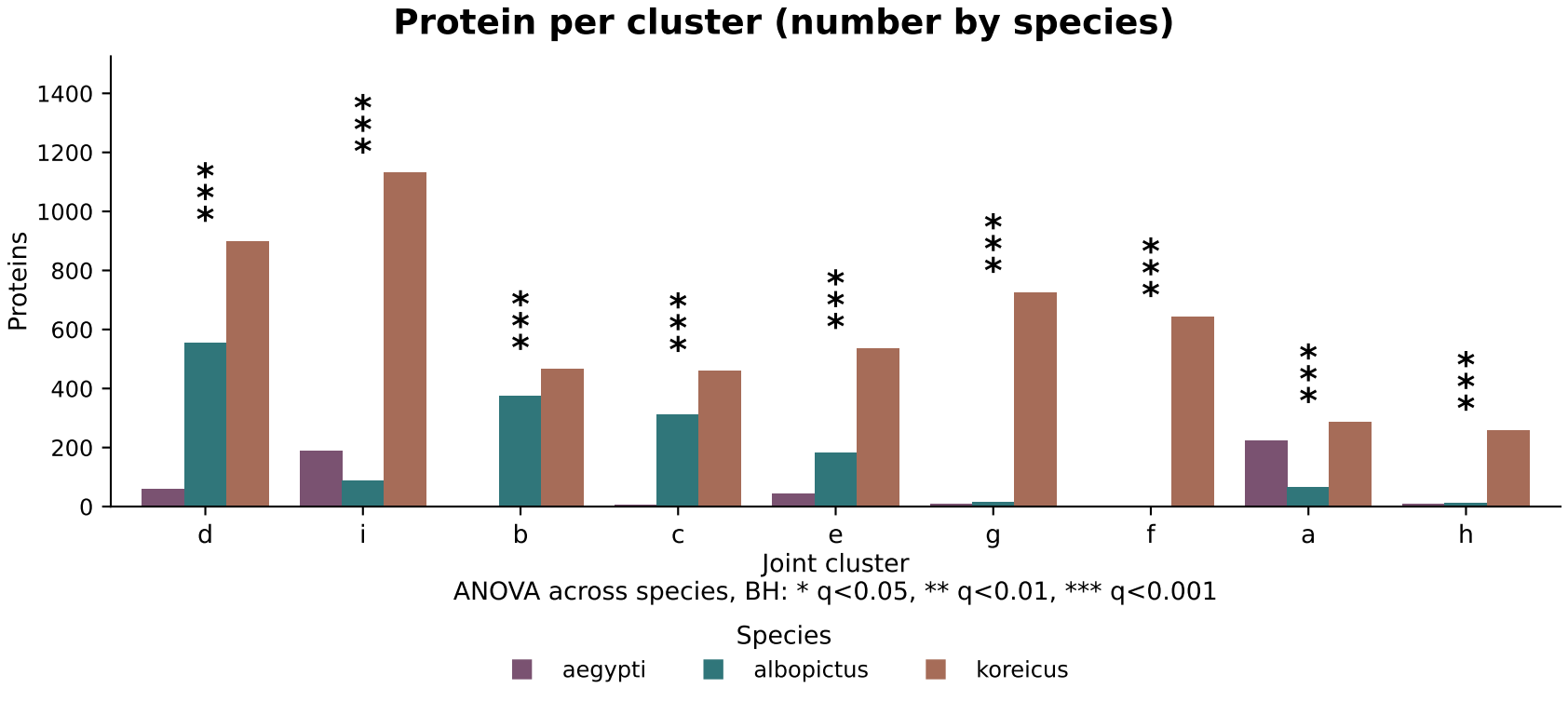

Intra-species (t-SNE)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
t-SNE 1
t-SNE 2
Proteins per highlighted cluster
500
1000
1500
2000
2500


Inter-species (UMAP)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
UMAP 1
UMAP 2
a
c
d
e
h
i
f
g
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Inter-species (t-SNE)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
t-SNE 1
t-SNE 2
a
b
c
d
e
f
g
h
i
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Top Inter-species clusters (re-projected)
aegypti (n=538)
albopictus (n=1,600)
koreicus (n=5,402)
UMAP 1
UMAP 2
Joint cluster
a
b
c
d
e
f
g
h
i


Intra- to Inter-species cluster fluxes
aegypti
(n=15,090, k=49)
albopictus
(n=13,096, k=38)
koreicus
(n=26,629, k=33)
10
11
a (223)
7
8
13
b (374)
14
c (312)
15
d (555)
17
e (181)
19
1
5
6
7
8
9
f (644)
g (726)
11
12
h (259)
13
14
i (1,130)
15
e (536)
16
17
18
d (899)
21
Cluster Specificity Index (S_i); black border: $S_i \geq 0.80$
0.0
0.2
0.4
0.6
0.8
1.0


Protein per cluster (number by species)
Proteins
1400
1200
1000
800
600
400
200
0
****
****
****
****
****
****
****
****
****
d
i
b
c
e
g
f
a
h
Joint cluster
ANOVA across species, BH: * q<0.05, ** q<0.01, *** q<0.001
Species
aegypti
albopictus
koreicus